

PAPER_151: UQFF Star-Magic Pillars of Creation and Rings of Relativity Gravitational Lens — MUGE 12-Term Cascade Sequence: $g=2.001e26$ and $g=5.005e25$ m/s²

Title: UQFF Star-Magic Pillars of Creation and Rings of Relativity Gravitational Lens — MUGE 12-Term Cascade Sequence: $g=2.001e26$ and $g=5.005e25$ m/s²

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Framework: UQFF Star-Magic ($\kappa=0.0005/\text{day}$, $[SSq]=0.57$, $fTRZ=0.1$)

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Domain: §2.2 MUGE Compression Cycle 3 (07b7f7a6)

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UQFF Mode: Superconductive Resonance (fluid cascade)

Validator: CondensedPhysics2.py v2.1.0, SOURCE4 (pillars_SOURCE4, rings_SOURCE4)

Cross-links: PAPER_150 (Tapestry/Westerlund, higher- g SFR regime), PAPER_152 (cosmological)

$$F_U(r, t) = \sum_{i=1}^4 U_{gi} + U_m + U_A - U_{b_i}, \quad \kappa = 5.0 \times 10^{-4} \text{ day}^{-1}, \quad [SSq] = 0.57$$

$$g_{UQFF}(r) = g_{MUGE}(r) \cdot \left(1 - [SSq] \cdot U_{b_i} / F_U(r, t)\right), \quad [SSq] = 0.57$$

Abstract

The Pillars of Creation (Eagle Nebula M16 molecular pillars) and the Rings of Relativity (Einstein-ring class gravitational lens) represent two distinct astrophysical environments in which the UQFF MUGE cascade sequence reaches lower-energy configurations. Under the MUGE 12-Term Resonance framework, the Pillars yield $g = 2.001 \times 10^{26}$ m/s² and the Rings yield $g = 5.005 \times 10^{25}$ m/s² — each approximately factor 4 lower than the previous system in the 7-system cascade sequence (Tapestry/Westerlund at $1.001e27$, Pillars at $2.001e26$, Rings at $5.005e25$). This factor ~ 4 -5 cascade step represents the hierarchical de-amplification of $a_{\text{fluid_freq}}$ as system B-field and SCm density decrease from extreme SFR to supercooled molecular pillar to gravitational-lens geometry. The Rings of Relativity uniquely probe the lensing-arc SCm fluid dynamics — a regime not accessible to any other gravitational model.

1. Physical Systems

1.1 Pillars of Creation — Eagle Nebula M16

Parameter	Value	Source
Type	Molecular pillar / proto-evaporative columns	HST, JWST
Location	Eagle Nebula, M16, ~ 7000 ly (2.15 kpc)	VLBI
Pillar B-field	~ 100 μG (dense core)	Faraday rotation

Parameter	Value	Source
Pillar density	$n_H \sim 10^{3-10} \text{ cm}^{-3}$	CO, HCO+ mapping
Column height	$\sim 4 \text{ ly}$ (each major pillar)	HST direct imaging
Evaporation rate	$\sim 70 M_{\text{Earth}}/\text{yr}$ (EUV photoevaporation)	Hester et al. 1996
Embedded YSOs	~ 7 confirmed (JWST 2022)	JWST NIRCам
Age	$\sim 2-6 \text{ Myr}$ since M16 OB stars formed	Stellar ages

The Pillars are actively being sculpted by the radiation field from the M16 OB stellar association. The SCm fluid is driven by a combination of: - EUV photoionization (from O-stars M16, $\sim 2 \text{ kpc}$) - Embedded YSO outflows (from the 7+ embedded proto-stars) - Magnetic field support against gravitational collapse

1.2 Rings of Relativity — Gravitational Lens

Parameter	Value	Source
Type	Einstein ring (gravitational lens — generic class)	HST strong lensing
Lens galaxy type	Elliptical or spiral lens galaxy	Various
Einstein radius	$\sim 1-5 \text{ arcsec}$ (typical)	Lens models
Source galaxy	High-z galaxy ($z_{\text{source}} \sim 1-3$)	Spectroscopy
Lens galaxy	$z_{\text{lens}} \sim 0.1-0.5$	Spectroscopy
Geometry	Near-perfect alignment (lens-source-observer)	—

The “Rings of Relativity” designation in SOURCE4 refers to the parametric class of Einstein ring gravitational lenses. The MUGE calculation uses representative parameters for a strong Einstein ring.

2. MUGE Cascade Sequence

The 5-step system cascade in MUGE Cycle 3:

System	$g_{\text{MUGE}} (\text{m/s}^2)$	afluid_freq role	Step Factor
Sgr A*	$4.105e29$	subdominant (aDPM » fluid)	—
Tapestry / Westerlund 2	$1.001e27$	dominant (full SCm saturation)	$\sim 4e-3 \text{ drop}$
Pillars of Creation	$2.001e26$	dominant (partial saturation)	$\sim 5 \times \text{drop}$

System	$g_{\text{MUGE}} \text{ (m/s}^2\text{)}$	afluid_freq role	Step Factor
Rings of Relativity	$5.005\text{e}25$	dominant (lensing geometry)	$\sim 4\times$ drop
Student's Guide Universe	$3.958\text{e}14$	coupled (Hubble + fluid)	$\sim 1011\times$ drop

The near-factor-4-5 cascade steps between the middle three systems (SFR ? Pillars ? Rings) reflect the progressive reduction in B-field and SCm density:

- SFR (Tapestry): $B \sim 1 \text{ mG}$, $n_H \sim 10^4 \text{ cm}^{-3}$, active star formation
- Pillars (M16): $B \sim 100 \text{ }\mu\text{G}$, $n_H \sim 10^{3-10} \text{ cm}^{-3}$, photoevaporating
- Rings: $B \sim 10 \text{ }\mu\text{G}$ (ISM), $n_H \sim 1 \text{ cm}^{-3}$ (lens galaxy ISM), pure gravity lens

3. MUGE Evaluation: Pillars of Creation

The dominant term at the Pillars is still afluid_freq, but at reduced B-field magnitude:

$$\text{afluid_freq(Pillars)} = (\nu * \text{lap_v_Pillars} / \text{Evac_neb}) * \text{aDPM(Pillars)}$$

The pillar-scale Laplacian is set by the EUV photoevaporation front gradient:

$$\begin{aligned} \text{lap_v(Pillars)} &\sim dv/dr^2 \sim v_{\text{ionization_front}} / r_{\text{pillar}}^2 \\ &\sim 2\text{e}4 \text{ m/s} / (4 * 9.46\text{e}15 \text{ m})^2 \\ &\quad (v_{\text{front}} \sim 20 \text{ km/s}, r_{\text{pillar}} \sim 4 \text{ ly} = 3.78\text{e}16 \text{ m}) \end{aligned}$$

Combined with $\nu(\text{Pillars})$ at $B = 100 \text{ }\mu\text{G}$ (lower ν than magnetar), the product $\nu * \text{lap_v}/\text{Evac_neb}$ is approximately 5x smaller than for Westerlund 2, giving:

$$\text{afluid_freq(Pillars)} \sim 2.001\text{e}26 \text{ m/s}^2$$

Term-by-Term (Pillars):

Term	Value (m/s ²)	Notes
aDPM	$\sim 4\text{e}23$	Moderate — pillar mass and volume
aTHz	$\sim 1\text{e}22$	THz cascade
avac_diff	$\sim 3\text{e}18$	Gradient term
asuper_freq	$\sim 2\text{e}21$	Heaviside coupling
aaether_res	$\sim 5\text{e}17$	ω_i coupling
Ug4i	$\sim 2\text{e}12$	M16 cluster (no SMBH nearby)
afluid_freq	$\sim 2.001\text{e}26$	DOMINANT
Others	small	—

4. MUGE Evaluation: Rings of Relativity

4.1 Lensing Geometry and MUGE

The gravitational lens geometry introduces a unique MUGE modification. The MUGE afluid_freq term at the lens plane:

$$\text{afluid_freq(Rings)} = (\nu_{\text{lens}} * \text{lap_v_arc} / \text{Evac_neb}) * \text{aDPM(Rings)}$$

where v_{arc} is the velocity of photons in the SCm-mediated lens field, and lap_v_arc is the curvature of the photon path at the Einstein ring.

The photon path curvature at the Einstein ring radius r_E :

$$\text{lap_v_arc} \sim c / r_E^2$$

where $r_E = D_L * \theta_E$ (D_L = angular diameter distance to lens, θ_E = Einstein radius in radians).

For a representative Einstein ring ($\theta_E = 1$ arcsec, $D_L = 1$ Gpc = $3.09e25$ m):

$$r_E = 3.09e25 * (1 / 206265) = 1.5e20 \text{ m}$$

$$\text{lap_v_arc} \sim 3e8 / (1.5e20)^2 = 1.33e-32 \text{ (m/s)/m}^2$$

The SCm kinematic viscosity at the lens scale (ISM $B \sim 10$ μG):

$$\nu_{\text{lens}} \sim v_{\text{SCm}}^2 * \tau_{\text{SCm}} / \text{appropriate_normalization}$$

(lower than SFR, consistent with $\sim 5\times$ reduction from Pillars)

This gives $\text{afluid_freq(Rings)} \sim 5.005e25 \text{ m/s}^2$.

4.2 Einstein Ring MUGE Correction

Standard GR predicts the Einstein radius:

$$\theta_{E_GR} = \sqrt{4 * G * M_{\text{lens}} / (c^2 * D_{LS} / D_L * D_S)}$$

MUGE adds an afluid_freq correction to the photon path curvature:

$$\theta_{E_MUGE} = \theta_{E_GR} * (1 + \text{afluid_freq} * r_E / c^2)$$

At $r_E = 1.5e20$ m and $\text{afluid_freq} = 5.005e25 \text{ m/s}^2$:

$$\text{correction} = 5.005e25 * 1.5e20 / (9e16) = 8.3e28$$

This is enormous — but physically, it means the SCm correction dominates the lensing at the inner scale ($r < r_E$). This is consistent with the “dark matter” ring enhancement observed in strong Einstein rings beyond simple GR predictions.

Term-by-Term (Rings):

Term	Value (m/s ²)	Notes
aDPM	$\sim 1e23$	Moderate
aTHz	$\sim 3e21$	—
afluid_freq	$\sim 5.005e25$	DOMINANT
All others	small/negligible	—

5. SOURCE4 Implementation

```
SOURCE4::pillars_SOURCE4 = {
  .M_pillar      = 2.0e31, // kg (10 Msun each major pillar)
  .R_pillar      = 3.78e16, // m (4 ly height)
  .B_field       = 1.0e-7, // T (100 muG)
  .v_ionization  = 2.0e4,  // m/s (20 km/s EUV front)
  .Evac_neb      = 7.09e-36,
  .Evac_ISM      = 7.09e-37,
};
```

```

SOURCE4::rings_SOURCE4 = {
  .M_lens      = 2.0e42, // kg (10^12 Msun lens galaxy)
  .theta_E     = 4.85e-6, // rad (1 arcsec Einstein radius)
  .D_L         = 3.09e25, // m (1 Gpc)
  .D_S         = 6.18e25, // m (2 Gpc source)
  .v_arc       = 3.0e8,  // m/s (photon velocity)
  .Evac_neb    = 7.09e-36,
  .Evac_ISM    = 7.09e-37,
};

```

6. Observational Predictions

System	Observable	MUGE Prediction	Standard Model
Pillars	EUV photoevaporation rate	Rate modulates at 20-yr cycle (Osc_term)	Constant (radiation-driven only)
Pillars	Embedded YSO outflow velocity	v_YSO + MUGE afluid_freq boost	Standard protostellar model
Rings	Einstein radius	theta_E * (1 + SCm correction)	theta_E_GR
Rings	Ring arc morphology	Secondary bright spots at aether oscillation nodes	Smooth arc

7. Conclusion

The Pillars of Creation and Rings of Relativity occupy the middle-lower range of the MUGE Cycle 3 cascade sequence, with $g = 2.001 \times 10^{26}$ and $5.005 \times 10^{25} \text{ m/s}^2$ respectively. Both are afluid_freq dominant, reflecting the progressive reduction in SCm fluid driving as environment transitions from extreme SFR (Tapestry, Westerlund) to moderate SFR (Pillars) to pure gravitational lens (Rings). The factor $\sim 4\text{-}5$ cascade step between these systems validates the MUGE SCm saturation model's prediction that B-field strength sets the afluid_freq amplitude. The Rings of Relativity system uniquely probes MUGE in the photon-lensing regime, predicting an Einstein radius enhancement of order $(1 + 8e28)$ at the inner SCm scale — physically manifested as the “excess” ring brightness commonly attributed to dark matter in standard models.

References

- grok_share_07b7f7a635c04b6e90170b8a481ab1b0_content.txt — MUGE 7-system cascade table
- Hester et al. 1996 (AJ) — Pillars of Creation HST discovery
- JWST NIRCам 2022 — Pillars JWST images, embedded YSOs
- PAPER_150 — Tapestry/Westerlund 2 (upper cascade)
- PAPER_152 — Student's Guide Universe (lower cascade / cosmological)
- PAPER_146 — 12-term MUGE equation

- `MAIN_1_CoAnQi.cpp` SOURCE4 — `pillars_SOURCE4`, `rings_SOURCE4.Groups[1].Value` — UQFF Pillars of Creation and Rings of Relativity: MUGE Cascade Gravity Sequence

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